

Functions: Composite, Inverse & Logarithmic

ANSWER KEY



Domain and range

- 1 Find the domain of $f(x) = \frac{1}{x^2 - 9}$.

The denominator is zero when $x = \pm 3$. Domain: all real x except $x = 3$ and $x = -3$.

- 2 Find the domain of $g(x) = \sqrt{7 - 2x}$.

The radicand must be non-negative: $7 - 2x \geq 0 \Rightarrow x \leq \frac{7}{2}$.

- 3 Find the range of $f(x) = -(x - 2)^2 + 5$.

Since $-(x - 2)^2 \leq 0$ for all x , $f(x) \leq 5$. Range: $f(x) \leq 5$.

Composite functions

- 4 Given $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $(f \circ g)(x)$.

$$f(g(x)) = (2x - 3)^2 + 1 = 4x^2 - 12x + 9 + 1 = 4x^2 - 12x + 10.$$

- 5 Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(x)$.

$$f(g(x)) = \sqrt{(x^2 - 4) + 4} = \sqrt{x^2} = |x|.$$

- 6 Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.

$$g(f(x)) = (\sqrt{x + 4})^2 - 4 = (x + 4) - 4 = x, \text{ valid for } x \geq -4 \text{ (the domain of } f).$$

Inverse functions

- 7 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 5}{3}$.

$$y = \frac{2x - 5}{3} \Rightarrow 3y = 2x - 5 \Rightarrow x = \frac{3y + 5}{2}. \text{ So } f^{-1}(x) = \frac{3x + 5}{2}.$$

- 8 Find $f^{-1}(x)$ for $f(x) = x^3 + 2$.

$$y = x^3 + 2 \Rightarrow x = \sqrt[3]{y - 2}. \text{ So } f^{-1}(x) = \sqrt[3]{x - 2}.$$

Verifying inverses

- 9 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses by showing $f(g(x)) = x$.

$$f(g(x)) = 5\left(\frac{x-1}{5}\right) + 1 = (x-1) + 1 = x. \text{ Confirmed.}$$

- 10 Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.

No. f is not one-to-one — it fails the horizontal line test, since for example $f(2) = f(-2) = 4$. An inverse function only exists if the domain is restricted (e.g. to $x \geq 0$).

Logarithm laws

- 11 Expand $\log_b \left(\frac{x^3 y}{z^2} \right)$ using the laws of logarithms.

$$\log_b \left(\frac{x^3 y}{z^2} \right) = 3\log_b x + \log_b y - 2\log_b z.$$

- 12 Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.

$$2\log_5 x + 3\log_5 y - \log_5 z = \log_5 \left(\frac{x^2 y^3}{z} \right).$$

- 13 Evaluate $\log_2 32 - \log_2 4$ without a calculator.

$$\log_2 \left(\frac{32}{4} \right) = \log_2 8 = 3.$$

Solving logarithmic equations

- 14 Solve $\log_4 (x^2) = 3$.

$$x^2 = 4^3 = 64 \Rightarrow x = \pm 8. \text{ Both satisfy the domain requirement } x^2 > 0. \text{ Solutions: } x = 8 \text{ or } x = -8.$$

- 15 Solve $\ln x + \ln(x-3) = \ln 10$.

$$\ln [x(x-3)] = \ln 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x-5)(x+2) = 0 \Rightarrow x = 5 \text{ or } x = -2. \text{ Since } \ln(x-3) \text{ requires } x > 3, \text{ reject } x = -2. \text{ Solution: } x = 5.$$

Exponential equations and logs as inverses

- 16 Solve $e^{2x-1} = 20$, giving your answer to three decimal places.

$$2x - 1 = \ln 20 \approx 2.996 \Rightarrow x = \frac{1 + 2.996}{2} \approx 1.998.$$

- 17 The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

$$81 = 3^4, \text{ so } x = 4. \text{ Since } f \text{ and } g \text{ are inverses, } g(f(16)) = 16.$$

Transformations of functions

- 18 The graph shows $f(x) = x^2$ (teal) and $g(x) = (x - 2)^2 + 1$ (indigo). Describe the transformation that maps f onto g .

g is f shifted right 2 units and up 1 unit — a horizontal translation of +2 followed by a vertical translation of +1.

- 19 If $h(x) = -f(x + 3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h .

Shift f left 3 units (from $x + 3$), then reflect the result over the x -axis (from the negative sign outside).

A well-designed word problem: investment growth and doubling time

- 20 This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t , to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.

(a) $A(10) = 8000e^{0.45} \approx 8000(1.5683) \approx 12546$ dollars. (b) Doubling requires $e^{0.045t} = 2 \Rightarrow 0.045t = \ln 2 \Rightarrow t = \frac{\ln 2}{0.045} \approx 15.4$ years. (c) From $A = A_0 e^{rt}$: $\frac{A}{A_0} = e^{rt} \Rightarrow \ln\left(\frac{A}{A_0}\right) = rt \Rightarrow t(A) = \frac{1}{0.045} \ln\left(\frac{A}{A_0}\right)$.

Multiple choice — easy

- 21 If $f(x) = -4x + 7$ and $g(x) = -2x - 5$, find $(f \circ g)(7)$.

a. 83 ✓

b. 85

c. -19

d. 81

$g(7) = -2(7) - 5 = -19$. $f(-19) = -4(-19) + 7 = 83$.

22 If $f(x) = -x + 2$ and $g(x) = -4x + 3$, find $(f \circ g)(-1)$.

a. -5 ✓

b. 7

c. -3

d. -7

$$g(-1) = -4(-1) + 3 = 7. f(7) = -(7) + 2 = -5.$$

23 If $f(x) = x$ and $g(x) = -3x + 3$, find $(f \circ g)(6)$.

a. -13

b. -15 ✓

c. -12

d. -17

$$g(6) = -3(6) + 3 = -15. f(-15) = -15.$$

24 If $f(x) = x + 7$ and $g(x) = -3x + 9$, find $(f \circ g)(6)$.

a. 0

b. -9

c. -4

d. -2 ✓

$$g(6) = -3(6) + 9 = -9. f(-9) = (-9) + 7 = -2.$$

25 If $f(x) = 2x - 6$ and $g(x) = -3x + 1$, find $(f \circ g)(3)$.

a. -20

b. -22 ✓

c. -8

d. -24

$$g(3) = -3(3) + 1 = -8. f(-8) = 2(-8) - 6 = -22.$$

26 If $f(x) = -3x + 8$ and $g(x) = -4x + 7$, find $(f \circ g)(8)$.

a. 85

b. 81

c. -25

d. 83 ✓

$$g(8) = -4(8) + 7 = -25. f(-25) = -3(-25) + 8 = 83.$$

27 If $f(x) = x$ and $g(x) = 3x - 9$, find $(f \circ g)(4)$.

a. 6

b. 5

c. 1

d. 3 ✓

$$g(4) = 3(4) - 9 = 3. f(3) = 3.$$

28 If $f(x) = 2x + 1$ and $g(x) = -2x - 7$, find $(f \circ g)(4)$.

a. -15

b. -29 ✓

c. -31

d. -27

$$g(4) = -2(4) - 7 = -15. f(-15) = 2(-15) + 1 = -29.$$

29 If $f(x) = 2x + 7$ and $g(x) = 4x + 4$, find $(f \circ g)(5)$.

- a. 57
- b. 53
- c. 24
- d. 55 ✓

$$g(5) = 4(5) + 4 = 24. f(24) = 2(24) + 7 = 55.$$

30 If $f(x) = x - 1$ and $g(x) = 2x + 4$, find $(f \circ g)(-4)$.

- a. -3
- b. -7
- c. -4
- d. -5 ✓

$$g(-4) = 2(-4) + 4 = -4. f(-4) = (-4) - 1 = -5.$$

31 If $f(x) = x - 8$ and $g(x) = -4x - 8$, find $(f \circ g)(1)$.

- a. -12
- b. -18
- c. -22
- d. -20 ✓

$$g(1) = -4(1) - 8 = -12. f(-12) = (-12) - 8 = -20.$$

32 If $f(x) = 4x + 8$ and $g(x) = 2x - 2$, find $(f \circ g)(-7)$.

- a. -56 ✓
- b. -16
- c. -58
- d. -54

$$g(-7) = 2(-7) - 2 = -16. f(-16) = 4(-16) + 8 = -56.$$

33 If $f(x) = 4x - 6$ and $g(x) = -2x - 6$, find $(f \circ g)(-5)$.

a. 8

b. 12

c. 10 ✓

d. 4

$$g(-5) = -2(-5) - 6 = 4. f(4) = 4(4) - 6 = 10.$$

34 If $f(x) = -4x + 5$ and $g(x) = 2x + 3$, find $(f \circ g)(-2)$.

a. 11

b. 7

c. -1

d. 9 ✓

$$g(-2) = 2(-2) + 3 = -1. f(-1) = -4(-1) + 5 = 9.$$

35 If $f(x) = 2x - 3$ and $g(x) = -4x + 2$, find $(f \circ g)(6)$.

a. -49

b. -45

c. -22

d. -47 ✓

$$g(6) = -4(6) + 2 = -22. f(-22) = 2(-22) - 3 = -47.$$

Multiple choice — medium

36 Find $f^{-1}(x)$ for $f(x) = \frac{6x-5}{8}$.

a. $\frac{6x+5}{8}$

b. $\frac{8x+5}{6}$ ✓

c. $\frac{8x-5}{6}$

d. $\frac{6x-5}{8}$

$$y = \frac{6x-5}{8} \Rightarrow 8y = 6x-5 \Rightarrow x = \frac{8y+5}{6}. \text{ So } f^{-1}(x) = \frac{8x+5}{6}.$$

37 Find $f^{-1}(x)$ for $f(x) = \frac{2x-16}{5}$.

a. $\frac{2x-16}{5}$

b. $\frac{5x-16}{2}$

c. $\frac{5x+16}{2}$ ✓

d. $\frac{2x+16}{5}$

$$y = \frac{2x-16}{5} \Rightarrow 5y = 2x-16 \Rightarrow x = \frac{5y+16}{2}. \text{ So } f^{-1}(x) = \frac{5x+16}{2}.$$

38 Find $f^{-1}(x)$ for $f(x) = \frac{8x-1}{5}$.

a. $\frac{5x-1}{8}$

b. $\frac{8x+1}{5}$

c. $\frac{5x+1}{8}$ ✓

d. $\frac{8x-1}{5}$

$$y = \frac{8x-1}{5} \Rightarrow 5y = 8x-1 \Rightarrow x = \frac{5y+1}{8}. \text{ So } f^{-1}(x) = \frac{5x+1}{8}.$$

39 Find $f^{-1}(x)$ for $f(x) = \frac{9x-13}{4}$.

a. $\frac{9x-13}{4}$

b. $\frac{4x-13}{9}$

c. $\frac{9x+13}{4}$

d. $\frac{4x+13}{9}$ ✓

$$y = \frac{9x-13}{4} \Rightarrow 4y = 9x-13 \Rightarrow x = \frac{4y+13}{9}. \text{ So } f^{-1}(x) = \frac{4x+13}{9}.$$

40 Find $f^{-1}(x)$ for $f(x) = \frac{3x - 18}{2}$.

a. $\frac{2x + 18}{3}$ ✓

b. $\frac{3x - 18}{2}$

c. $\frac{2x - 18}{3}$

d. $\frac{3x + 18}{2}$

$$y = \frac{3x - 18}{2} \Rightarrow 2y = 3x - 18 \Rightarrow x = \frac{2y + 18}{3}. \text{ So } f^{-1}(x) = \frac{2x + 18}{3}.$$

41 Find $f^{-1}(x)$ for $f(x) = \frac{5x - 14}{3}$.

a. $\frac{5x - 14}{3}$

b. $\frac{3x - 14}{5}$

c. $\frac{3x + 14}{5}$ ✓

d. $\frac{5x + 14}{3}$

$$y = \frac{5x - 14}{3} \Rightarrow 3y = 5x - 14 \Rightarrow x = \frac{3y + 14}{5}. \text{ So } f^{-1}(x) = \frac{3x + 14}{5}.$$

42 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 10}{7}$.

a. $\frac{7x + 10}{2}$ ✓

b. $\frac{2x - 10}{7}$

c. $\frac{7x - 10}{2}$

d. $\frac{2x + 10}{7}$

$$y = \frac{2x - 10}{7} \Rightarrow 7y = 2x - 10 \Rightarrow x = \frac{7y + 10}{2}. \text{ So } f^{-1}(x) = \frac{7x + 10}{2}.$$

43 Find $f^{-1}(x)$ for $f(x) = \frac{7x - 15}{4}$.

a. $\frac{4x + 15}{7}$ ✓

b. $\frac{7x - 15}{4}$

c. $\frac{7x + 15}{4}$

d. $\frac{4x - 15}{7}$

$$y = \frac{7x - 15}{4} \Rightarrow 4y = 7x - 15 \Rightarrow x = \frac{4y + 15}{7}. \text{ So } f^{-1}(x) = \frac{4x + 15}{7}.$$

44 Find $f^{-1}(x)$ for $f(x) = \frac{6x - 4}{4}$.

a. $\frac{6x - 4}{4}$

b. $\frac{4x - 4}{6}$

c. $\frac{4x + 4}{6}$ ✓

d. $\frac{6x + 4}{4}$

$$y = \frac{6x - 4}{4} \Rightarrow 4y = 6x - 4 \Rightarrow x = \frac{4y + 4}{6}. \text{ So } f^{-1}(x) = \frac{4x + 4}{6}.$$

45 Find $f^{-1}(x)$ for $f(x) = \frac{6x - 18}{4}$.

a. $\frac{4x + 18}{6}$ ✓

b. $\frac{4x - 18}{6}$

c. $\frac{6x - 18}{4}$

d. $\frac{6x + 18}{4}$

$$y = \frac{6x - 18}{4} \Rightarrow 4y = 6x - 18 \Rightarrow x = \frac{4y + 18}{6}. \text{ So } f^{-1}(x) = \frac{4x + 18}{6}.$$

46 Find $f^{-1}(x)$ for $f(x) = \frac{5x - 2}{8}$.

a. $\frac{5x - 2}{8}$

b. $\frac{8x - 2}{5}$

c. $\frac{5x + 2}{8}$

d. $\frac{8x + 2}{5}$ ✓

$$y = \frac{5x - 2}{8} \Rightarrow 8y = 5x - 2 \Rightarrow x = \frac{8y + 2}{5}. \text{ So } f^{-1}(x) = \frac{8x + 2}{5}.$$

47 Find $f^{-1}(x)$ for $f(x) = \frac{4x - 20}{2}$.

a. $\frac{2x - 20}{4}$

b. $\frac{2x + 20}{4}$ ✓

c. $\frac{4x - 20}{2}$

d. $\frac{4x + 20}{2}$

$$y = \frac{4x - 20}{2} \Rightarrow 2y = 4x - 20 \Rightarrow x = \frac{2y + 20}{4}. \text{ So } f^{-1}(x) = \frac{2x + 20}{4}.$$

48 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 11}{6}$.

a. $\frac{2x - 11}{6}$

b. $\frac{2x + 11}{6}$

c. $\frac{6x - 11}{2}$

d. $\frac{6x + 11}{2}$ ✓

$$y = \frac{2x - 11}{6} \Rightarrow 6y = 2x - 11 \Rightarrow x = \frac{6y + 11}{2}. \text{ So } f^{-1}(x) = \frac{6x + 11}{2}.$$

49 Find $f^{-1}(x)$ for $f(x) = \frac{8x - 2}{5}$.

a. $\frac{5x + 2}{8}$ ✓

b. $\frac{8x + 2}{5}$

c. $\frac{8x - 2}{5}$

d. $\frac{5x - 2}{8}$

$$y = \frac{8x - 2}{5} \Rightarrow 5y = 8x - 2 \Rightarrow x = \frac{5y + 2}{8}. \text{ So } f^{-1}(x) = \frac{5x + 2}{8}.$$

50 Find $f^{-1}(x)$ for $f(x) = \frac{6x-3}{3}$.

a. $\frac{6x-3}{3}$

b. $\frac{6x+3}{3}$

c. $\frac{3x-3}{6}$

d. $\frac{3x+3}{6}$ ✓

$$y = \frac{6x-3}{3} \Rightarrow 3y = 6x-3 \Rightarrow x = \frac{3y+3}{6}. \text{ So } f^{-1}(x) = \frac{3x+3}{6}.$$

Multiple choice — hard

51 Solve $\ln x + \ln(x-4) = \ln 140$.

a. 14 ✓

b. 12

c. 16

d. -10

$$\ln[x(x-4)] = \ln 140 \Rightarrow x(x-4) = 140 \Rightarrow x^2 - 4x - 140 = 0 \Rightarrow (x-14)(x-(-10)) = 0 \Rightarrow x = 14 \text{ (rejecting } x = -10 \text{ since } x > 4 \text{ is required)}.$$

52 Solve $\ln x + \ln(x-5) = \ln 6$.

a. 8

b. -1

c. 4

d. 6 ✓

$$\ln[x(x-5)] = \ln 6 \Rightarrow x(x-5) = 6 \Rightarrow x^2 - 5x - 6 = 0 \Rightarrow (x-6)(x-(-1)) = 0 \Rightarrow x = 6 \text{ (rejecting } x = -1 \text{ since } x > 5 \text{ is required)}.$$

53 Solve $\ln x + \ln(x - 4) = \ln 77$.

a. 11 ✓

b. -7

c. 9

d. 13

$\ln[x(x - 4)] = \ln 77 \Rightarrow x(x - 4) = 77 \Rightarrow x^2 - 4x - 77 = 0 \Rightarrow (x - 11)(x - (-7)) = 0 \Rightarrow x = 11$ (rejecting $x = -7$ since $x > 4$ is required).

54 Solve $\ln x + \ln(x - 4) = \ln 221$.

a. 15

b. 17 ✓

c. 19

d. -13

$\ln[x(x - 4)] = \ln 221 \Rightarrow x(x - 4) = 221 \Rightarrow x^2 - 4x - 221 = 0 \Rightarrow (x - 17)(x - (-13)) = 0 \Rightarrow x = 17$ (rejecting $x = -13$ since $x > 4$ is required).

55 Solve $\ln x + \ln(x - 6) = \ln 247$.

a. -13

b. 17

c. 21

d. 19 ✓

$\ln[x(x - 6)] = \ln 247 \Rightarrow x(x - 6) = 247 \Rightarrow x^2 - 6x - 247 = 0 \Rightarrow (x - 19)(x - (-13)) = 0 \Rightarrow x = 19$ (rejecting $x = -13$ since $x > 6$ is required).

56 Solve $e^{3x-9} = 433$, giving your answer to three decimal places.

- a. 5.524
- b. 4.524
- c. 6.024
- d. 5.024 ✓

$$3x - 9 = \ln 433 \approx 6.071 \Rightarrow x = \frac{9 + 6.071}{3} \approx 5.024.$$

57 Solve $e^{3x-7} = 105$, giving your answer to three decimal places.

- a. 3.385
- b. 3.885 ✓
- c. 4.885
- d. 4.385

$$3x - 7 = \ln 105 \approx 4.654 \Rightarrow x = \frac{7 + 4.654}{3} \approx 3.885.$$

58 Solve $e^{4x-9} = 67$, giving your answer to three decimal places.

- a. 2.801
- b. 3.301 ✓
- c. 3.801
- d. 4.301

$$4x - 9 = \ln 67 \approx 4.205 \Rightarrow x = \frac{9 + 4.205}{4} \approx 3.301.$$

59 Solve $e^{4x-6} = 488$, giving your answer to three decimal places.

- a. 4.048
- b. 3.548
- c. 3.048 ✓
- d. 2.548

$$4x - 6 = \ln 488 \approx 6.190 \Rightarrow x = \frac{6 + 6.190}{4} \approx 3.048.$$

60 Solve $e^{2x-8} = 355$, giving your answer to three decimal places.

a. 6.436

b. 7.936

c. 6.936 ✓

d. 7.436

$$2x - 8 = \ln 355 \approx 5.872 \Rightarrow x = \frac{8 + 5.872}{2} \approx 6.936.$$

61 Evaluate $\log_3 648 - \log_3 8$ without a calculator.

a. 3

b. 5

c. 4 ✓

d. 640

$$\log_3 \left(\frac{648}{8} \right) = \log_3 81 = 4.$$

62 Evaluate $\log_5 500 - \log_5 4$ without a calculator.

a. 496

b. 3 ✓

c. 4

d. 2

$$\log_5 \left(\frac{500}{4} \right) = \log_5 125 = 3.$$

63 Evaluate $\log_3 5103 - \log_3 7$ without a calculator.

a. 7

b. 5

c. 5096

d. 6 ✓

$$\log_3 \left(\frac{5103}{7} \right) = \log_3 729 = 6.$$

64 Evaluate $\log_3 72 - \log_3 8$ without a calculator.

a. 64

b. 2 ✓

c. 3

d. 1

$$\log_3 \left(\frac{72}{8} \right) = \log_3 9 = 2.$$

65 Evaluate $\log_2 32 - \log_2 2$ without a calculator.

a. 30

b. 3

c. 4 ✓

d. 5

$$\log_2 \left(\frac{32}{2} \right) = \log_2 16 = 4.$$